

Graphical Presentation

Contents

- ❑ **For qualitative data**

- ❑ Bar chart

- ❑ Pie chart

- ❑ **For quantitative data**

- ❑ Histogram

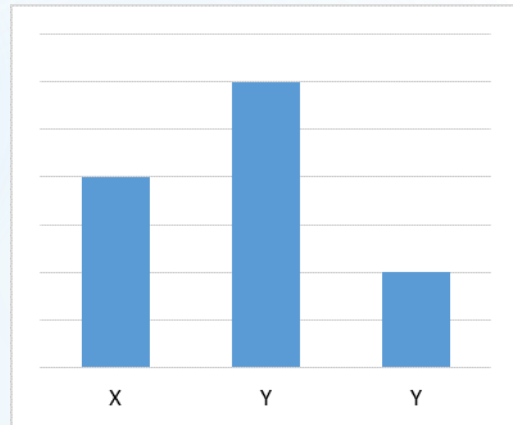
- ❑ Frequency polygon

- ❑ Cumulative frequency curve (Ogive)

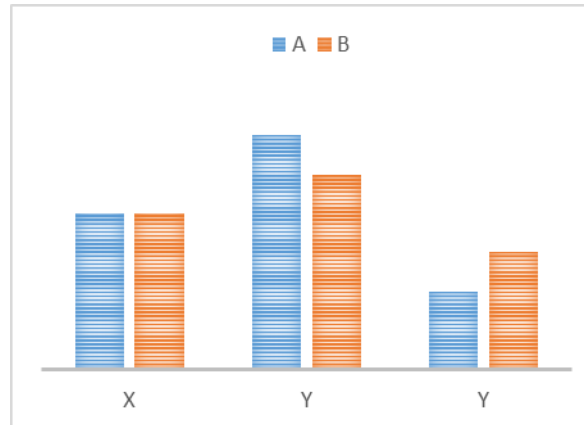
Bar Chart

3

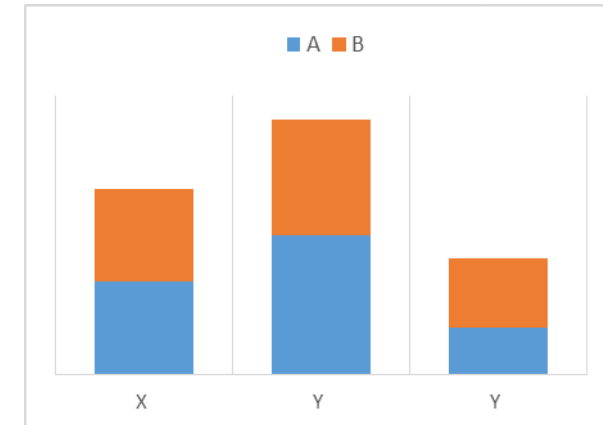
Simple bar chart



Clustered/ Multiple Bar chart



Stacked/ Component bar chart



Pie Chart

4

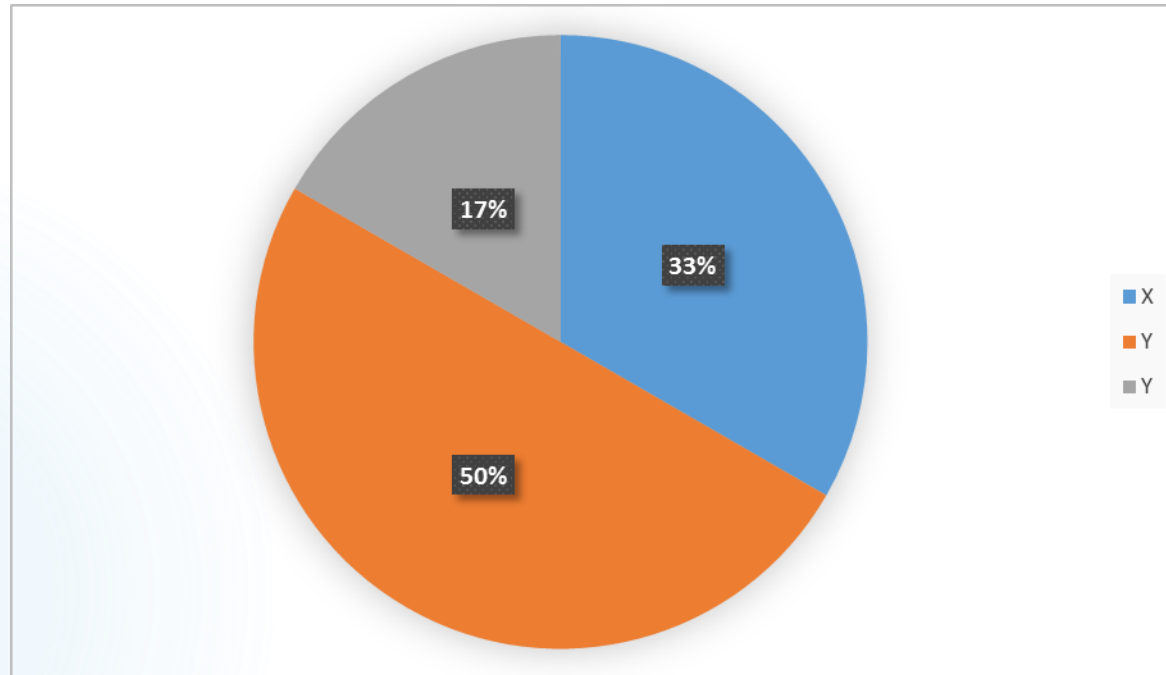


Chart (Diagram) Example

5

□ Example

The following is the frequency distribution table of highest education level of 145 randomly selected respondents-

Education level	No. of respondents (frequency)
No education	40
Primary	30
Secondary	25
Higher	50

Represent this in a bar and a pie chart

Chart (Diagram) Example

6

□ Example

The following is the frequency distribution table of highest education level of 145 randomly selected respondents-

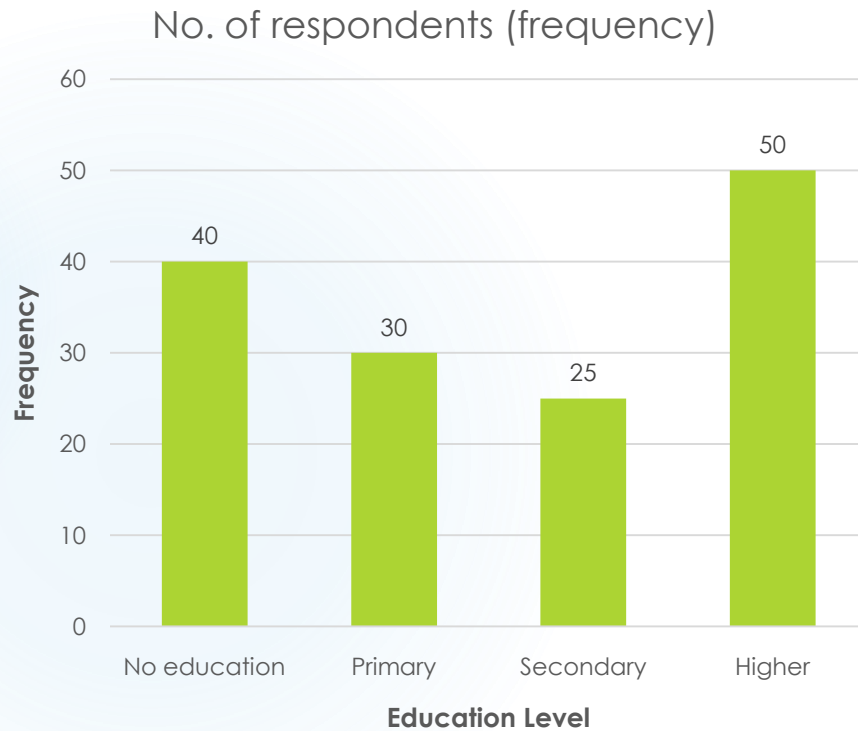
Education level	No. of respondents (frequency)	Percentages
No education	40	27.6
Primary	30	20.7
Secondary	25	17.2
Higher	50	34.5
Total	145	100

Represent this in a bar and a pie chart

Chart (Diagram) Example

7

Bar Chart



Pie Chart

No. of respondents (frequency)

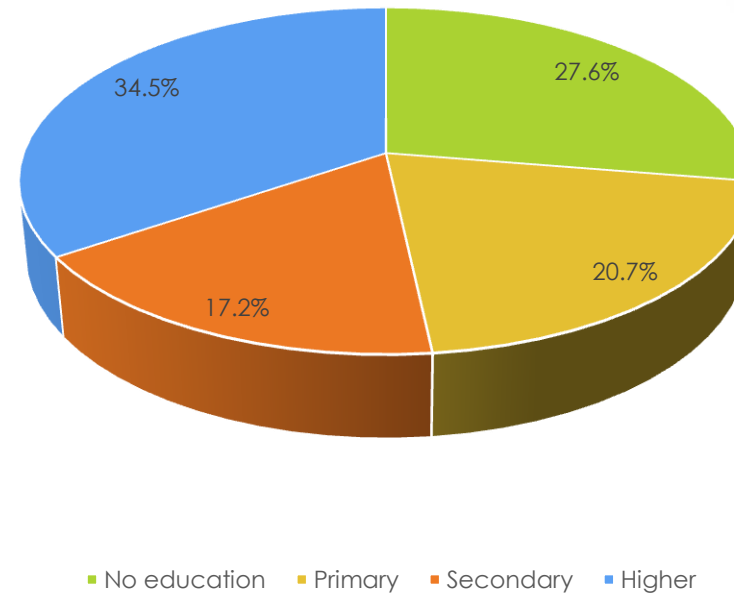


Chart (Diagram) Example

8

□ Example

The following is the Cross-tabulation of **Highest education level** by **Gender** of 145 randomly selected respondents-

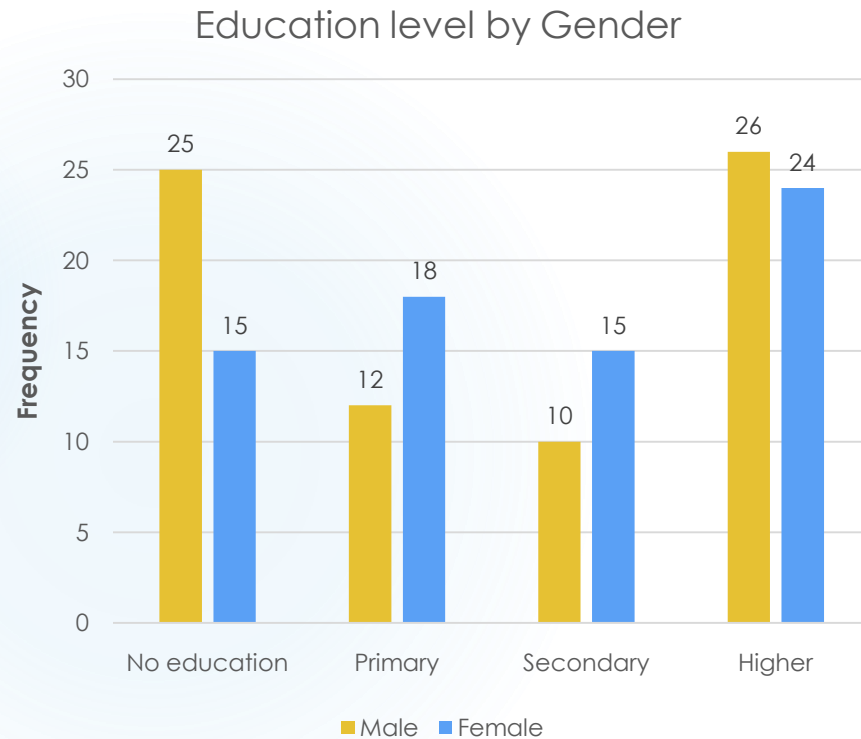
Education level	Gender		Total
	Male	Female	
No education	25	15	40
Primary	12	18	30
Secondary	10	15	25
Higher	26	24	50
Total	73	72	145

Represent this in a multiple bar chart and a component bar chart

Chart (Diagram) Example

9

Multiple Bar Chart



Component Bar Chart

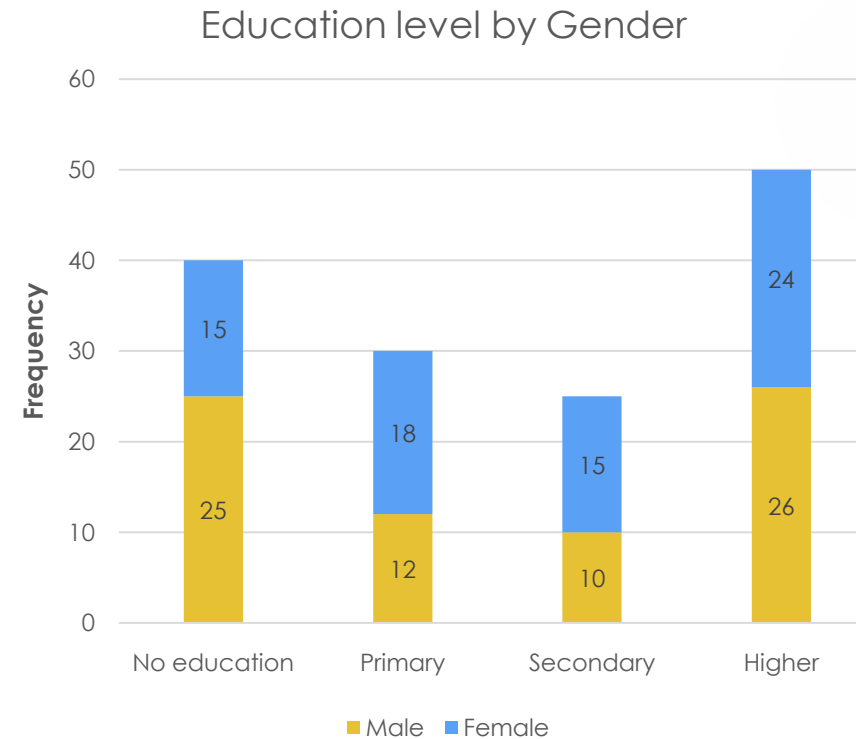
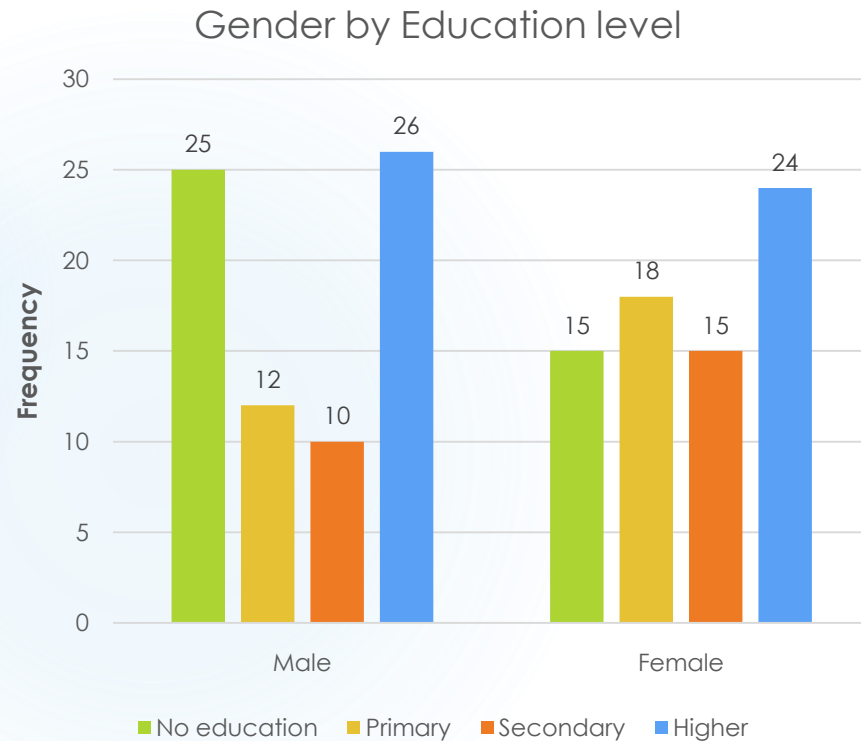


Chart (Diagram) Example

10

Multiple Bar Chart



Component Bar Chart

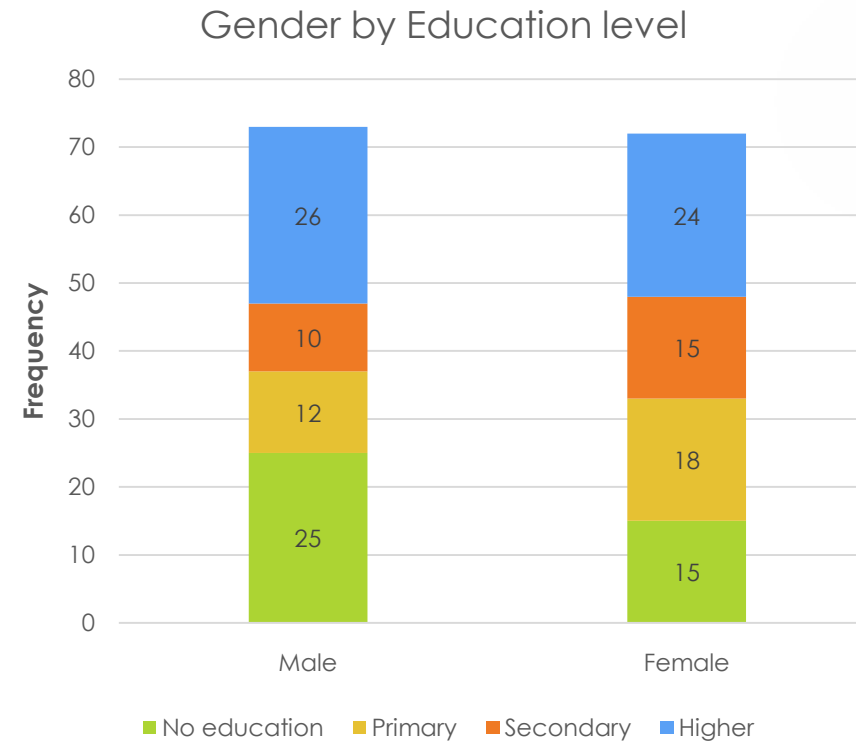


Chart (Diagram) Example

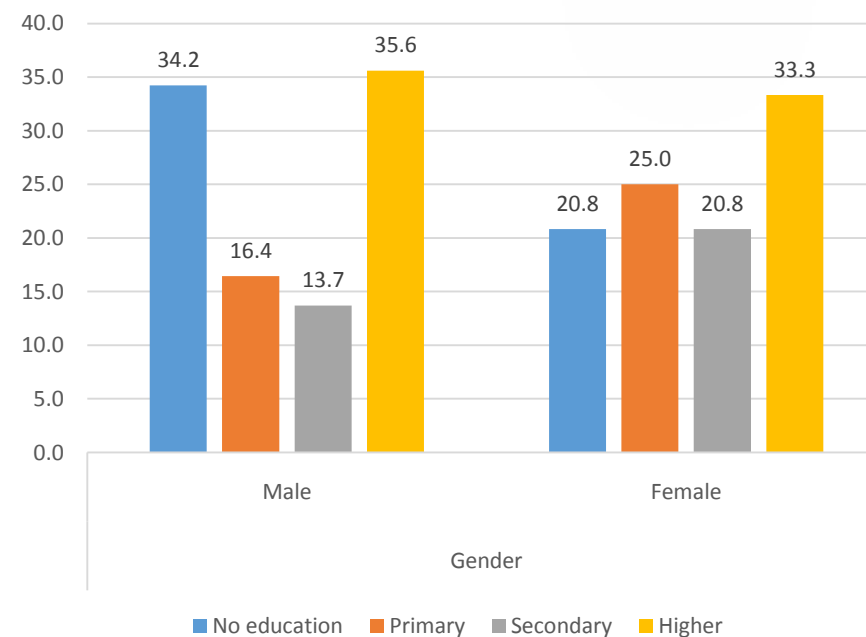
11

Multiple bar chart taking percentages for each x-axis category-

Education level	Gender	
	Male	Female
No education	= $25/73*100\%$ = 34.2%	= $15/72*100\%$ = 20.8%
Primary	= $12/73*100\%$ = 16.4%	= $18/72*100\%$ = 25.0%
Secondary	= $10/73*100\%$ = 13.7%	= $15/72*100\%$ = 20.8%
Higher	= $26/73*100\%$ = 35.6%	= $24/72*100\%$ = 33.3%
Total	73	72

Multiple Bar Chart

Percentage distribution for educational level
(Percentages calculated for each x-axis category)



Graphs (Plots)

12

□ Example

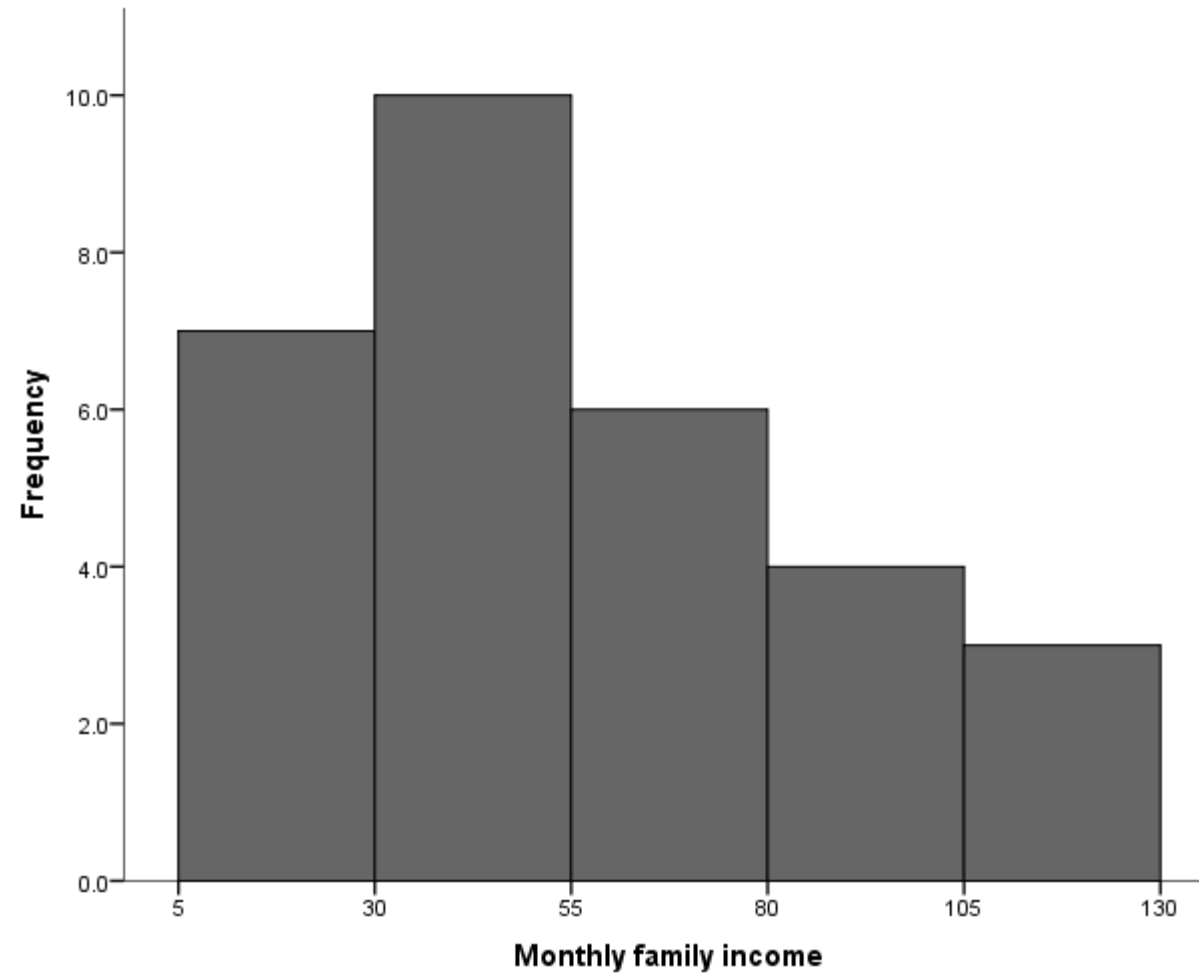
Below given a frequency distribution table of monthly family income of the respondent-

Monthly family income	No. of respondents
05-30	7
30-55	10
55-80	6
80-105	4
105-130	3

Show this in a histogram, a frequency polygon and an Ogive

Histogram

13



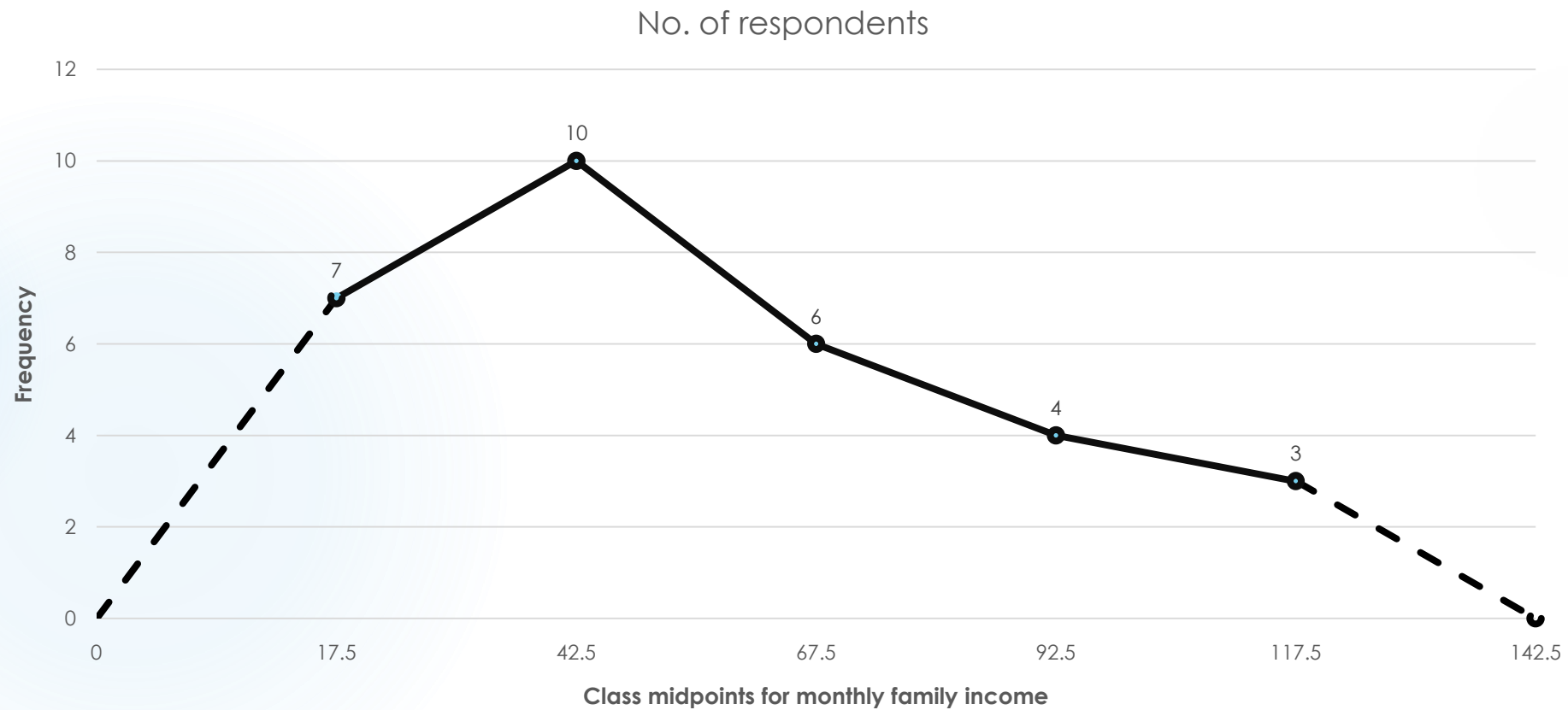
Frequency Polygon

14

Monthly family income	Class Midpoints	No. of respondents
05-30	17.5	7
30-55	42.5	10
55-80	67.5	6
80-105	92.5	4
105-130	117.5	3

Frequency Polygon

15



Cumulative Frequency Curve (Ogive)

16

Monthly family income	No. of respondents	Cumulative Frequency	
		Less than type	Greater than type
05-30	7	7	30
30-55	10	17	23
55-80	6	23	13
80-105	4	27	7
105-130	3	30	3

Cumulative Frequency Curve (Ogive)

17

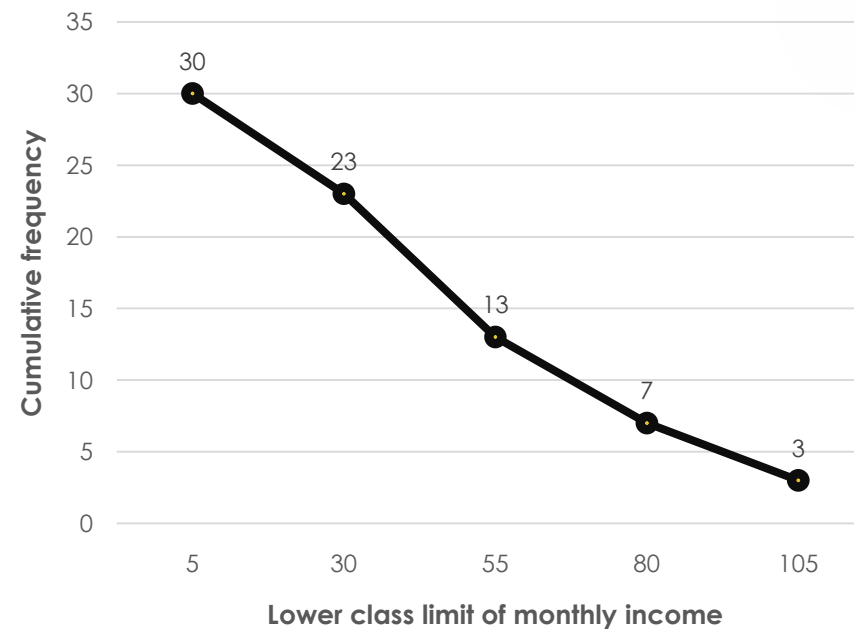
Less Than Type Ogive

Less than type Ogive for monthly family income



Greater Than type Ogive

Greater than type Ogive for monthly family income



Assignment

18

Question 1: Below given the frequency distribution of wealth status of the respondents of a sample of size 200.

Wealth Status	Frequency (Number of respondents)
Poorest	45+ last 1 digit of ID
Poor	35+ last 1 digit of ID
Middle	60+ last 1 digit of ID
Rich	40+ last 1 digit of ID
Richest	20+ last 1 digit of ID
Total	200

Draw a Bar chart showing the above information. What can you interpret from the chart?

Assignment

19

Question 2: Below given the age distribution of 150 respondents

Age Groups	Frequency (Number of respondents)
0-20	25+ last 1 digit of ID
20-40	35+ last 1 digit of ID
40-60	45+ last 1 digit of ID
60-80	30+ last 1 digit of ID
80-100	15+ last 1 digit of ID
Total	150

Draw a Histogram and a less than type Ogive. What can you say from these graphs?